Master Data Management (MDM) Plan- MediLink Health Network

5.1 Introduction

Master Data Management (MDM) is the process of creating, maintaining, and governing a consistent, trusted representation of critical business entities across an organisation. For MediLink Health Network, patient information is a core master data domain because the same patient may receive healthcare services at multiple facilities.

MediLink operates across Kampala, Mukono, Mbarara, and Gulu. Each facility may maintain its own patient records, resulting in duplicate registrations, inconsistent personal information, and conflicting identifiers. These problems can affect continuity of care, reporting accuracy, and the reliability of patient information.

This phase establishes an MDM framework to identify patient records, resolve duplicates, create a trusted golden patient record, and maintain links to the original records at each facility.

5.2 Objectives of MDM

The objectives of the MediLink MDM implementation are to:

1. Establish a unique enterprise identity for each verified patient.
2. Identify duplicate patient records across participating facilities.
3. Create a trusted golden patient record from validated source information.
4. Maintain links between the golden record and the original facility records.
5. Define rules for resolving conflicting patient information.
6. Establish data stewardship, ownership, and approval procedures.
7. Preserve audit trails for record matching, merging, and correction.

5.3 Master Data Management Architecture

MediLink will adopt a centralised, registry-based MDM approach with controlled golden-record creation. Source systems at individual facilities remain responsible for their operational records, while the central platform maintains enterprise patient identities and links to source records.

The process follows these stages:

1. Source registration: Receive patient records from participating facilities.
2. Standardisation: Clean and standardise identifiers, names, phone numbers, and dates.
3. Matching: Compare records using available identifiers and demographic attributes.
4. Match classification: Classify records as confirmed matches, possible matches, or distinct patients.

5. Stewardship: Refer ambiguous matches and conflicting information to an authorised data steward.
6. Golden-record creation: Establish a trusted consolidated patient record using approved survivorship rules.
7. Distribution and synchronisation: Make the enterprise identifier and approved patient information available to authorised systems.
8. Monitoring: Track changes, duplicate candidates, unresolved conflicts, and record quality.

This architecture supports centralised patient identification without requiring all healthcare facilities to abandon their existing operational systems.

5.4 Patient Identity Resolution

5.4.1 Standardisation

Before matching, patient records will be standardised to reduce false mismatches caused by differences in formatting.

Examples include:

- Removing unnecessary spaces from names.
- Standardising phone numbers to the agreed Ugandan international format, where valid.
- Converting dates into a consistent database representation.
- Normalising approved categorical values.
- Validating and comparing National ID hashes where available and lawfully processed.

Standardisation will not change the underlying meaning of a record or invent missing information.

5.4.2 Matching rules

MediLink will use a combination of deterministic and probabilistic matching.

Matching rule	Example	Decision
Exact enterprise identifier	Two records already reference the same verified enterprise patient ID.	Link to the same patient, subject to identity controls.
Exact National ID hash	Two records have the same verified National ID hash.	Strong match candidate, subject to validation and exception checks.
Exact phone number and date of birth	Two records share both attributes.	Strong candidate for further verification.

Similar name and matching date of birth	Names differ slightly, but dates of birth match.	Probabilistic candidate for review.
Name alone	Two patients have the same name.	Insufficient evidence to merge.
Conflicting identity attributes	Matching identifiers but materially conflicting demographic details.	Hold for data-steward investigation.

A shared phone number is not conclusive evidence of identity because family members may use the same phone. Likewise, missing identifiers must not automatically prevent a record from being considered for matching.

5.4.3 Match classification

Each comparison will be assigned a classification:

- Confirmed match: Sufficient evidence establishes that records represent the same patient.
- Possible match: Evidence suggests a match, but further verification is required.
- Non-match: Available evidence indicates distinct patients.
- Unresolved: Insufficient or conflicting evidence prevents a reliable decision.

Only authorised and sufficiently verified matches may proceed to consolidation.

5.5 Golden Patient Record

A golden record is the trusted enterprise representation of a patient, created from validated source records under approved matching and survivorship rules.

For MediLink, the golden patient record may contain:

Attribute	Purpose
Enterprise Patient ID	Unique identifier assigned by MediLink.
Full name	Best verified patient name.
Date of birth	Validated date of birth, with provenance.
Sex	Standardised value supported by source evidence.
Phone number	Best verified and current contact number.
National ID hash	Protected matching attribute, where available.
Record status	Indicates whether the enterprise record is active, under review, or otherwise restricted.
Source references	Links to contributing facility records.
Last verified date	Indicates when the relevant information was last checked.

The golden record will not simply select whichever value appears most frequently. Attribute-level survivorship rules will consider source reliability, recency, verification status, and the nature of the attribute.

For example, a recently verified phone number may be preferred over an older unverified number. A conflicting date of birth should be referred for investigation rather than silently overwritten.

5.6 Data Survivorship Rules

Data survivorship determines which value is retained when multiple source records provide different values for the same attribute.

Attribute	Survivorship rule
Patient name	Prefer a verified value; refer unresolved material conflicts to a data steward.
Date of birth	Prefer an authoritative, verified source; do not automatically overwrite conflicting values.
Phone number	Prefer the most recently verified valid number.
Sex	Retain the verified value supported by the source evidence and approved data standards.
National ID hash	Retain the validated hash with controlled access and appropriate provenance.
Facility association	Preserve all relevant source-facility relationships rather than selecting only one facility.

Every selected value should retain provenance indicating its source, verification status, and relevant update history. This allows authorised staff to investigate disagreements and understand how the golden record was produced.

5.7 Data Stewardship and Governance

MediLink will assign clear responsibilities for the management of master patient data.

Role	Responsibility
Data Governance Lead	Defines MDM policies, matching thresholds, and approval procedures.
Data Steward	Reviews possible matches, resolves conflicts, and approves authorised merges.

Data Custodian	Maintains source data, executes approved corrections, and monitors technical quality.
Database Administrator	Maintains database security, integrity constraints, availability, and backups.
Facility Data Officer	Verifies patient information and investigates errors originating at the facility.
Healthcare Personnel	Use approved patient identifiers and report suspected duplicate or incorrect records.

Patient-record merges and unmerges must be auditable. The system should preserve the original facility identifiers and retain sufficient information to reverse an incorrect merge without losing the underlying source history.

5.8 Implementation in MySQL 8.0+

The MDM implementation will use relational tables to maintain enterprise patient identities, source-system references, match candidates, and record-change history.

The core logical structures are:

- `master_patient`: Stores the enterprise patient identity and golden-record attributes.
- `patient_source_record`: Maps source patient identifiers to the corresponding enterprise patient.
- `patient_match_candidate`: Stores candidate matches and their review status.
- `mdm_merge_history`: Records approved merges, affected records, and the responsible user.
- `patient_attribute_history`: Records changes to important patient attributes and their provenance.

These structures describe the proposed MDM implementation. Their final names and columns must be aligned with the existing MediLink schema before SQL implementation.

MySQL constraints and transactions will protect relational integrity, while application-level matching logic or scheduled processing will identify candidate duplicates. Human review will remain necessary for ambiguous cases.

National ID hashes will be treated as sensitive matching data. A plain SHA-256 hash of a predictable identifier is not sufficient protection against guessing attacks by itself, so access restrictions and an appropriately designed keyed-hashing approach should be considered for production deployment.

5.9 MDM Performance Indicators

MediLink will monitor the effectiveness of its MDM implementation using the following indicators.

Indicator	Measurement
Duplicate detection rate	Confirmed duplicate cases detected ÷ all confirmed duplicate cases identified through validation.
Match-review completion	Candidate cases resolved or formally referred ÷ total candidate cases requiring review.
Golden-record completeness	Golden records containing all required attributes ÷ total eligible golden records.
Unresolved conflict rate	Golden-record candidates with unresolved material conflicts ÷ total records under review.
Incorrect merge rate	Confirmed incorrect merges ÷ total approved merges.
Source traceability	Golden records with valid links to contributing source records ÷ total golden records.

Targets will be approved by the data governance team after establishing baseline measurements. The incorrect merge rate should be minimised, while source traceability and review completion should approach 100%.